Scaling Multi-Document Event Summarization

Evaluating Compression vs. Full-Text Approaches

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NAACL 2025

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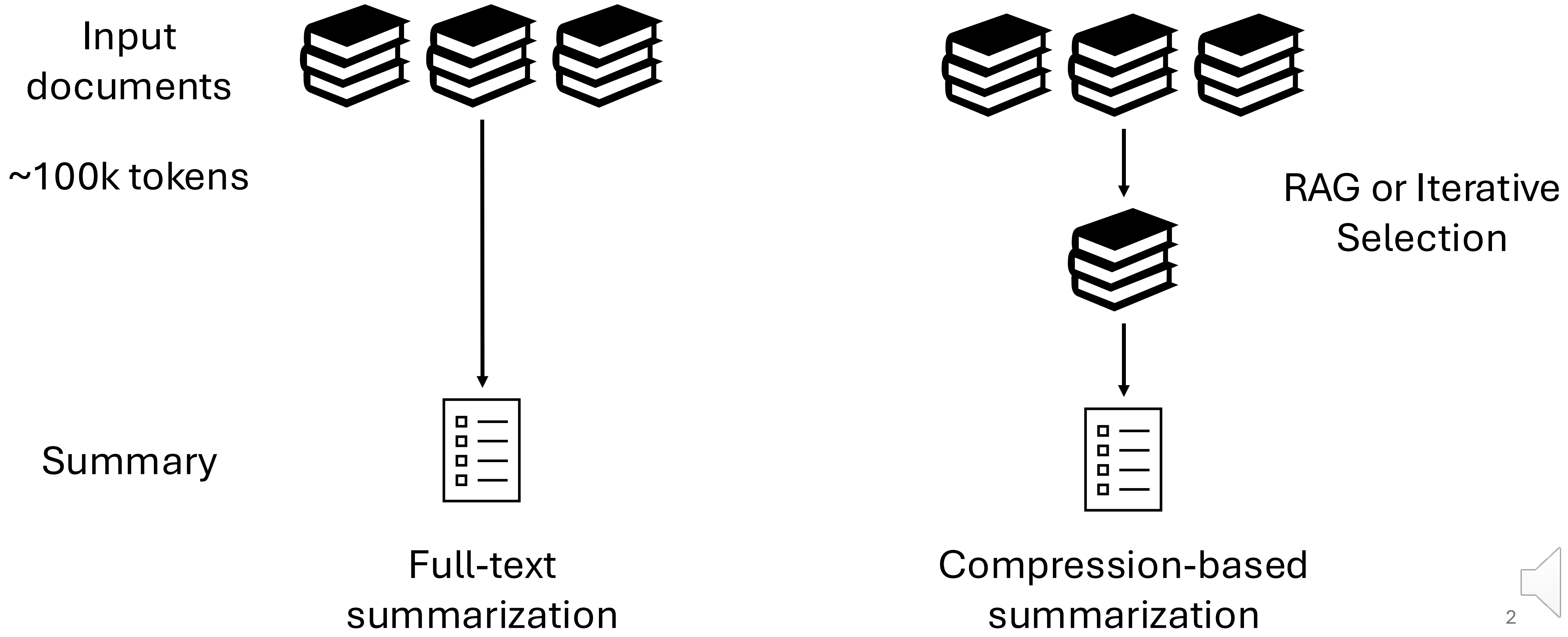
Task: Summarize 100 texts



Systems

- Models,
  - Llama 3.1 (8B, 70B)
  - Command-R (32B)
  - Jamba 1.5 (52B MoE, 12B active)
- All support at least 128k context window
- All models are competitive on long-context retrieval, RULER (Hsieh et al., 2024)

Hsieh et al., 2024. RULER: What's the Real Context Size of Your Long-Context Language Models? COLM

Compression methods

- Retrieval-augmented generation
  - SFR Embedding-2 to retrieve up to 32k tokens
- Iterative summarization (Chang et al., 2024)
  - Hierarchical
    - Summarize documents and iteratively summarize those summaries
  - Incremental
    - Summarize documents one-by-one in order, maintaining an overall summary

Chang et al., 2024. BooookScore: A systematic exploration of book-length summarization in the era of LLMs. ICLR

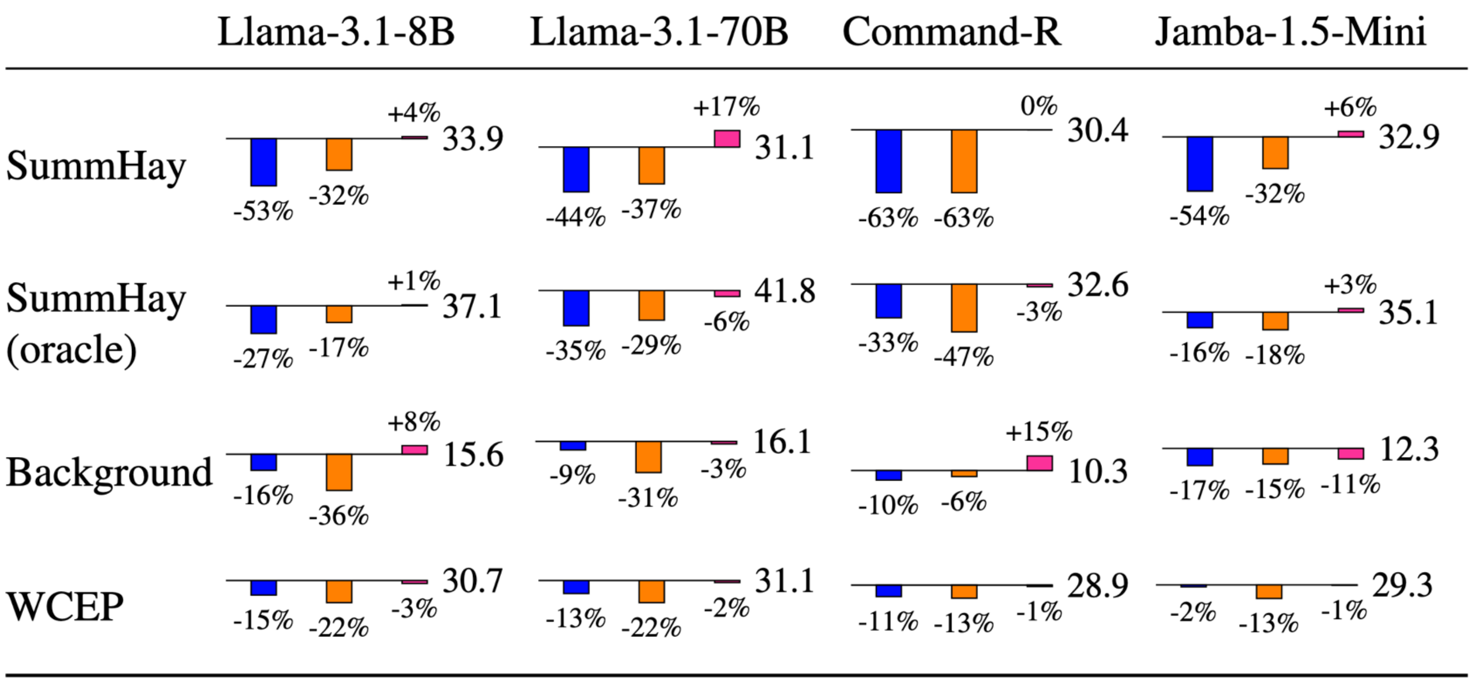
Datasets

- Summary of a Haystack (SummHay)
- Background
- Wikipedia Current Events Portal (WCEP)

			Avg. length	
Dataset	# Ex.	# Docs/Ex.	Doc.	Summ.
SummHay	92	100	884	185
Background	658	186	1033	174
WCEP	1020	76	468	34

- Laban et al., 2024. Summary of a Haystack: A Challenge to Long-Context LLMs and RAG Systems. EMNLP
- Pratapa et al., 2023. Background Summarization of Event Timelines. EMNLP
- Gholipour Ghalandari et al., 2020. A Large-Scale Multi-Document Summarization Dataset from the Wikipedia Current Events Portal. ACL

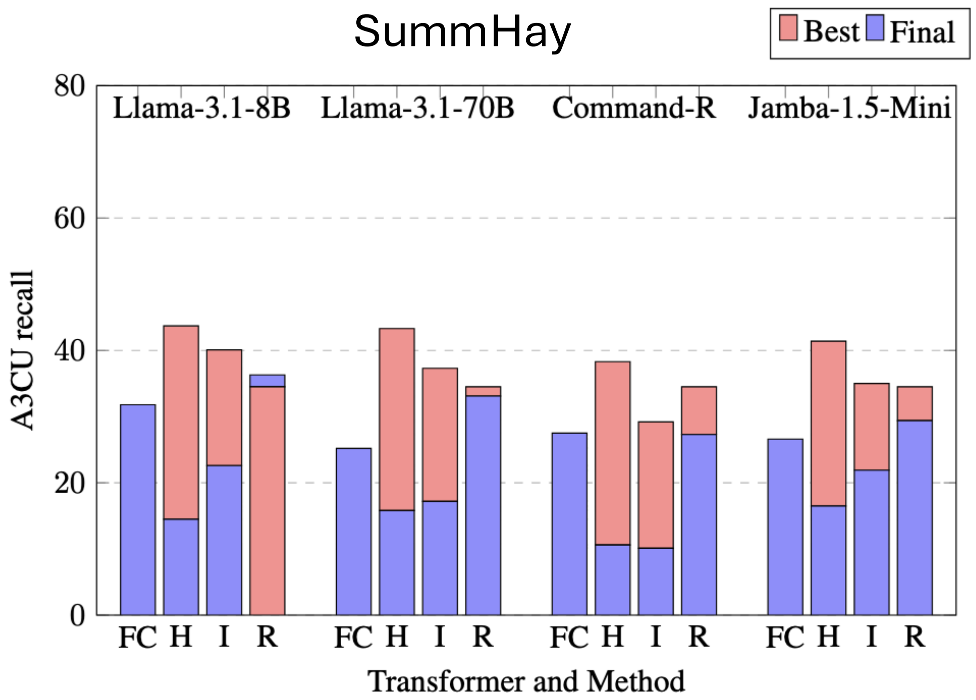
Overall Scores

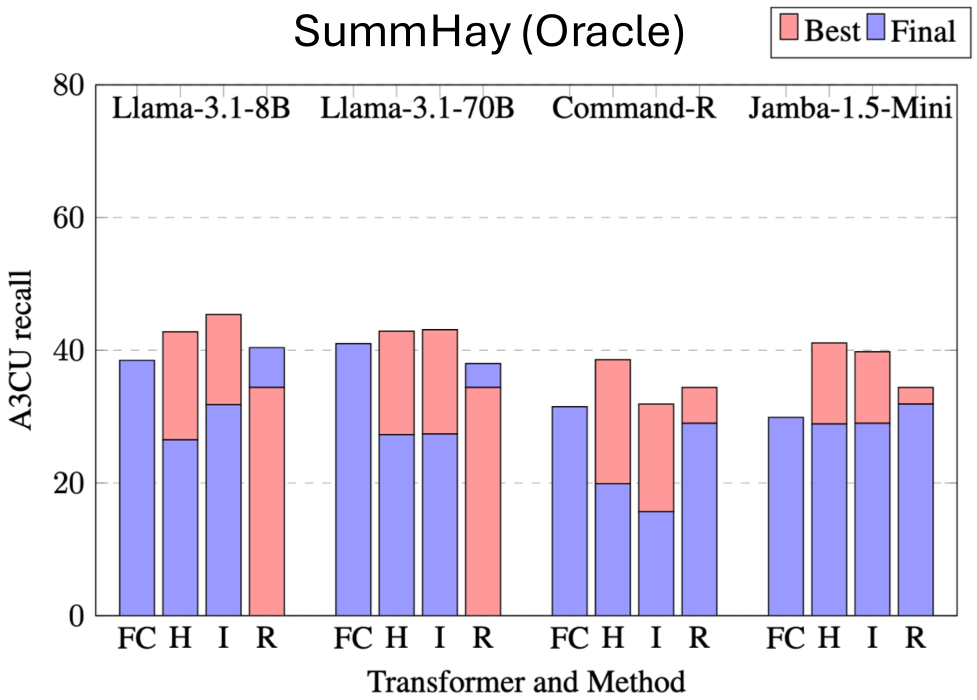


Relative performances of hierarchical, incremental and retrieval compared to full-text baseline (AutoACU F1)

Liu et al., 2023. Towards Interpretable and Efficient Automatic Reference-Based Summarization Evaluation. EMNLP

Analyzing Information Loss





We compare recall scores of final summary against the (best) intermediate outputs

What's next?

- Hybrid methods: Compression with a long-context window
  - Selective compression: LongLLMLingua, RAPTOR, RECOMP
  - Identifying optimal context window: RULER, HELMET
- Reference-free evaluation
  - A3CU relies on availability of human-written references